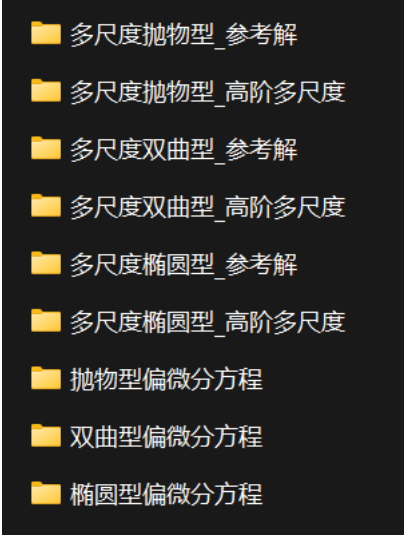


多尺度偏微分方程计算函数程序包使用示例

代码压缩包解压之后如图，

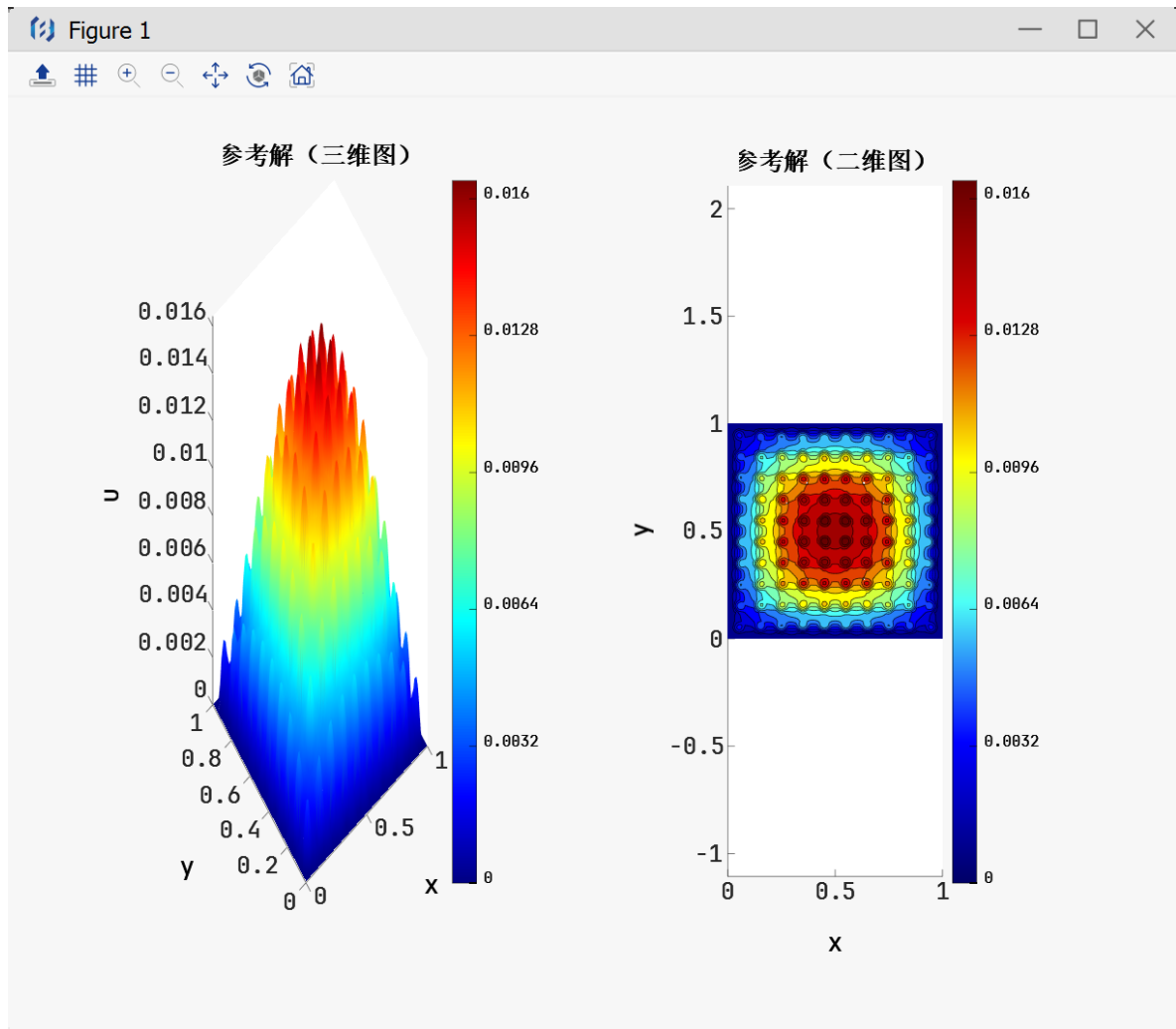


- 多尺度抛物型_参考解
- 多尺度抛物型_高阶多尺度
- 多尺度双曲型_参考解
- 多尺度双曲型_高阶多尺度
- 多尺度椭圆型_参考解
- 多尺度椭圆型_高阶多尺度
- 抛物型偏微分方程
- 双曲型偏微分方程
- 椭圆型偏微分方程

每个文件夹中都有关于该文件夹下主函数的使用示例，名称为 Example1.m。

一、多尺度椭圆型偏微分方程（参考解）

```
clc,clear;
%设定求解区域
xy_range=[0,1,0,1];
%设定差分网格
Nxy=[100,100];
%设定方程参数
equ_para.c_in=1;
equ_para.c_ma=100;
equ_para.epsilon=0.1;
equ_para.f=@(x,y)10;
equ_para.len=0.03;
%设定边界条件
bc_type=[1,1,1,1];
bc_para.h_left=@(x,y)1;
bc_para.r_left=@(x,y)0;
bc_para.h_right=@(x,y)1;
bc_para.r_right=@(x,y)0;
bc_para.h_bottom=@(x,y)1;
bc_para.r_bottom=@(x,y)0;
bc_para.h_top=@(x,y)1;
bc_para.r_top=@(x,y)0;
%设定求解方法
method='fivepoint';
solve_mulle1_dir(xy_range,Nxy,equ_para,bc_type,bc_para,method)
```



二、多尺度抛物型偏微分方程（参考解）

```

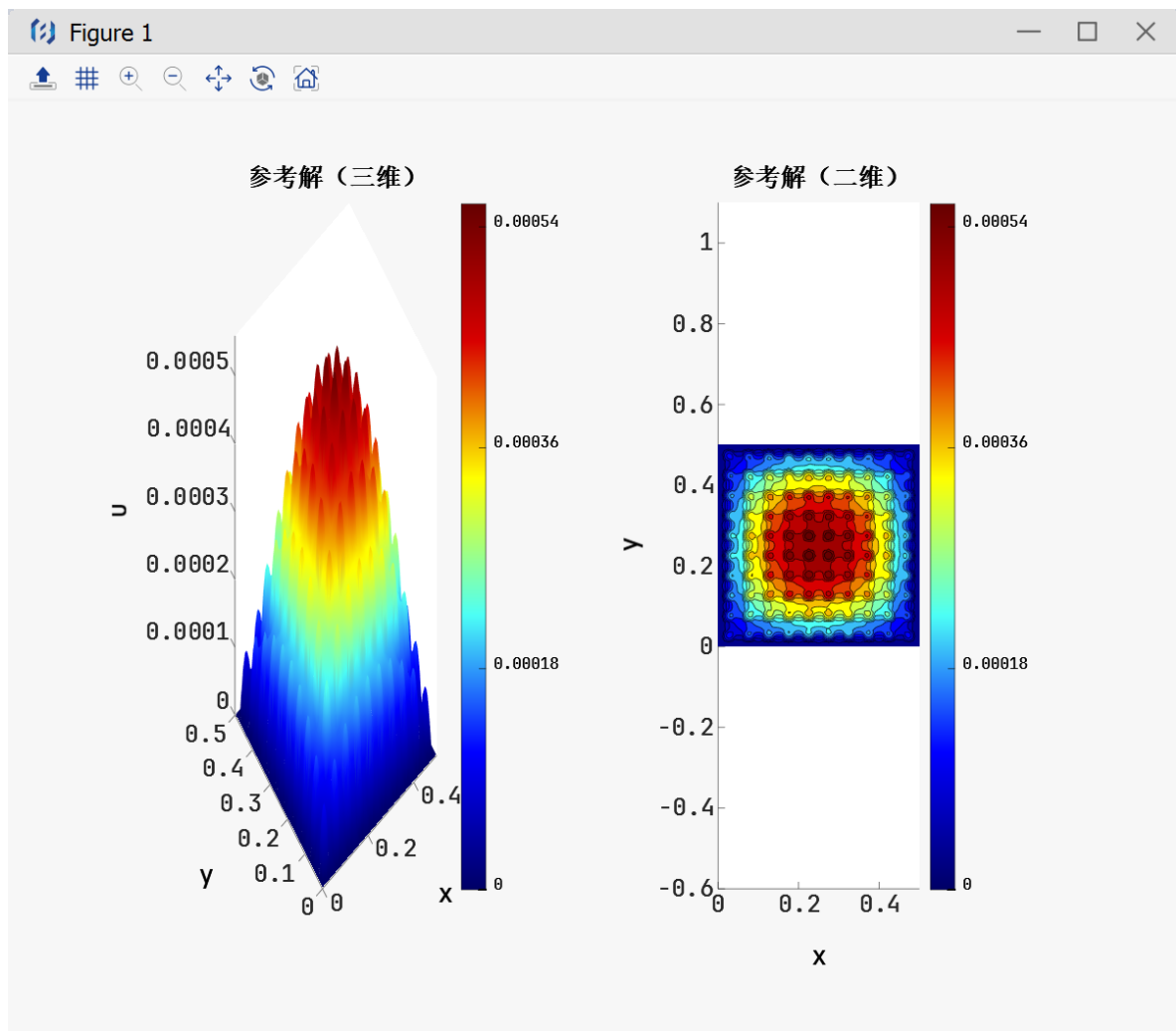
clc,clear;
%设定求解区域
xyt_range=[0,0.5,0,0.5,0,0.1];
%设定差分网格点数
Nxyt=[100,100,100];
%设定方程参数
equ_para.d_in=1;
equ_para.d_ma=1;
equ_para.c_in=1;
equ_para.c_ma=100;
equ_para.f=@(x,y,t)1;
equ_para.len=0.02;
equ_para.epsilon=0.1;
%设定初始条件
u0=@(x,y)0;
%设定边界条件
bc_type=[1,1,1,1];
bc_para.h_left=@(x,y,t)1;

```

```

bc_para.r_left=@(x,y,t)0;
bc_para.h_right=@(x,y,t)1;
bc_para.r_right=@(x,y,t)0;
bc_para.h_top=@(x,y,t)1;
bc_para.r_top=@(x,y,t)0;
bc_para.h_bottom=@(x,y,t)1;
bc_para.r_bottom=@(x,y,t)0;
solve_mulpdir(xyt_range,Nxyt,equ_para,u0,bc_type,bc_para)

```



三、多尺度双曲型偏微分方程（参考解）

```

clc,clear;
%设定求解区域
xyt_range=[0,1,0,1,0,0.1];
%设定差分网格
Nxyt=[200,200,2000];
%设定方程参数
equ_para.c_in=1;
equ_para.c_ma=100;
equ_para.d_in=1;

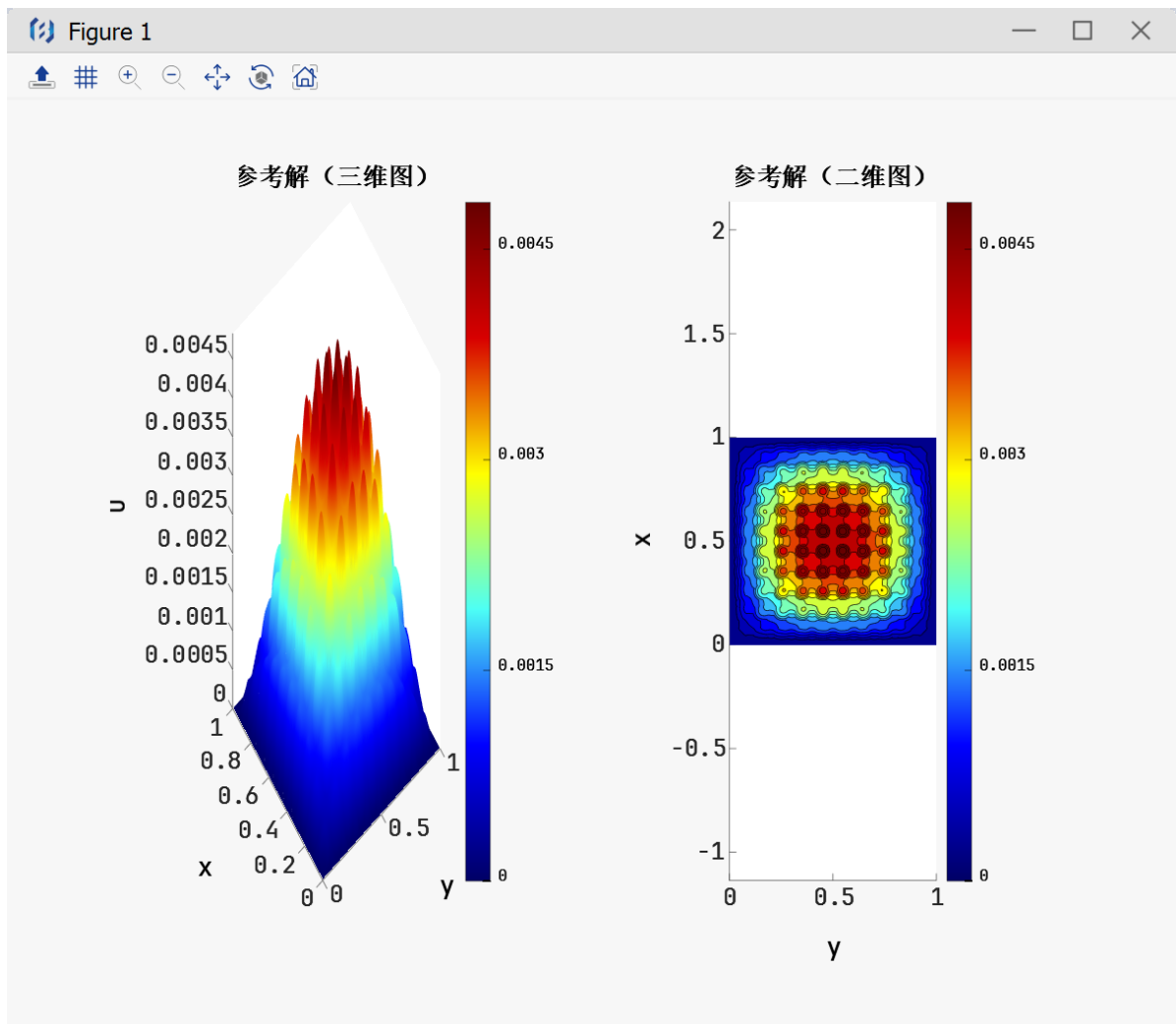
```

```

equ_para.d_ma=1;
equ_para.f=@(x,y,t)1;
equ_para.epsilon=0.1;
equ_para.len=0.04;
%设定初始条件
init.u0=@(x,y)0;
init.v0=@(x,y)0;
%设定边界条件
bc_type=[1,1,1,1];
bc_para.h_left=@(x,y,t)1;
bc_para.r_left=@(x,y,t)0;
bc_para.h_right=@(x,y,t)1;
bc_para.r_right=@(x,y,t)0;
bc_para.h_bottom=@(x,y,t)1;
bc_para.r_bottom=@(x,y,t)0;
bc_para.h_top=@(x,y,t)1;
bc_para.r_top=@(x,y,t)0;

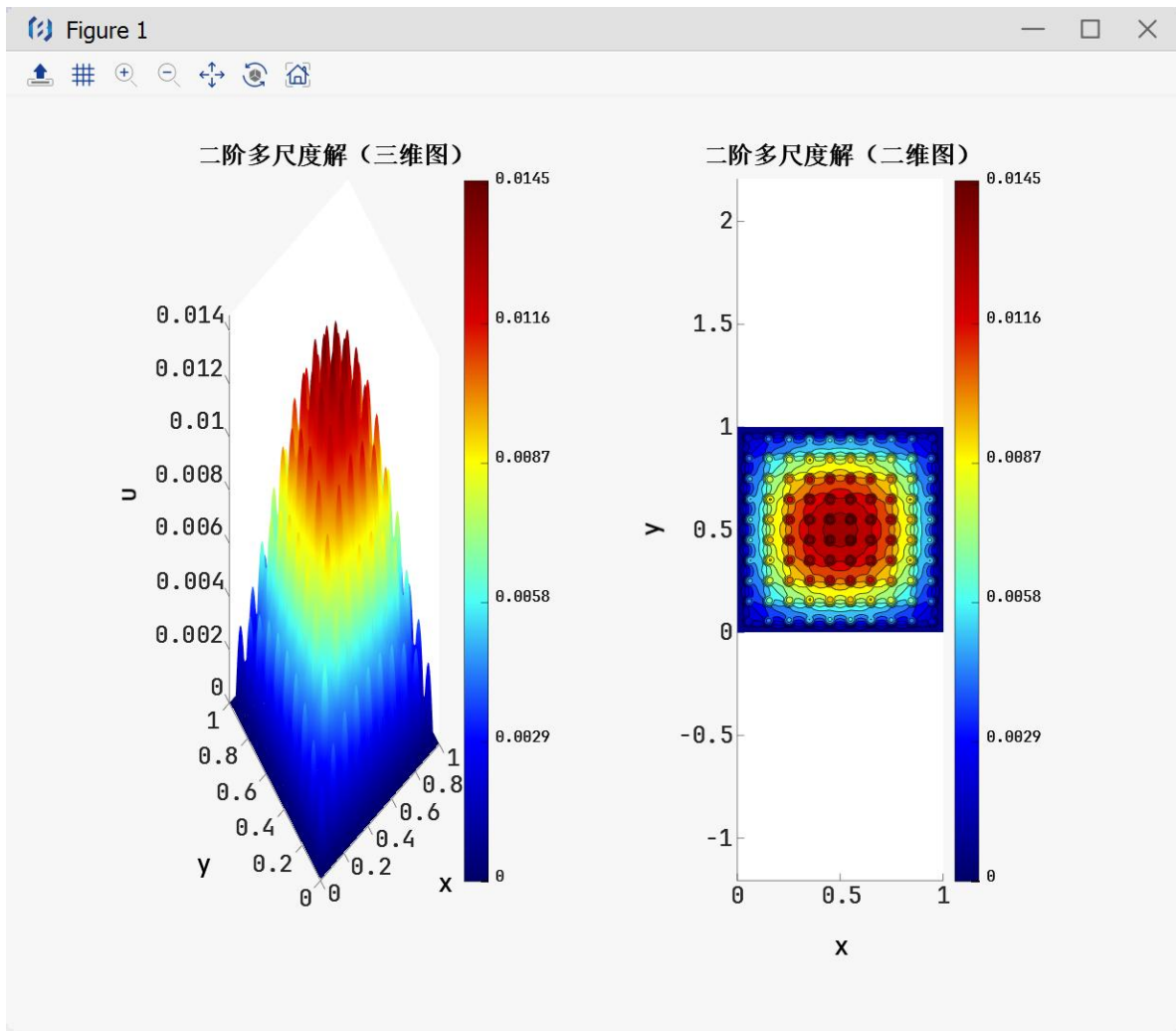
solve_mulhyp_dir(xyt_range,Nxyt,equ_para,init,bc_type,bc_para)

```



四、多尺度椭圆型偏微分方程（二阶多尺度解）

```
clc,clear;
%设定求解区域
xy_range=[0,1,0,1];
%设定差分网格
Nxy=[100,50,200];
%设定方程参数
equ_para.c_in=1;
equ_para.c_ma=100;
equ_para.epsilon=0.1;
equ_para.f=@(x,y)10;
equ_para.len=0.03;
%设定边界条件
bc_type=[1,1,1,1];
bc_para.h_left=@(x,y)1;
bc_para.r_left=@(x,y)0;
bc_para.h_right=@(x,y)1;
bc_para.r_right=@(x,y)0;
bc_para.h_bottom=@(x,y)1;
bc_para.r_bottom=@(x,y)0;
bc_para.h_top=@(x,y)1;
bc_para.r_top=@(x,y)0;
%设定夹杂相材料形状
shape='circle';
solve_mulle1(xy_range,Nxy,equ_para,bc_type,bc_para,shape)
```



五、多尺度抛物型偏微分方程（二阶多尺度解）

```

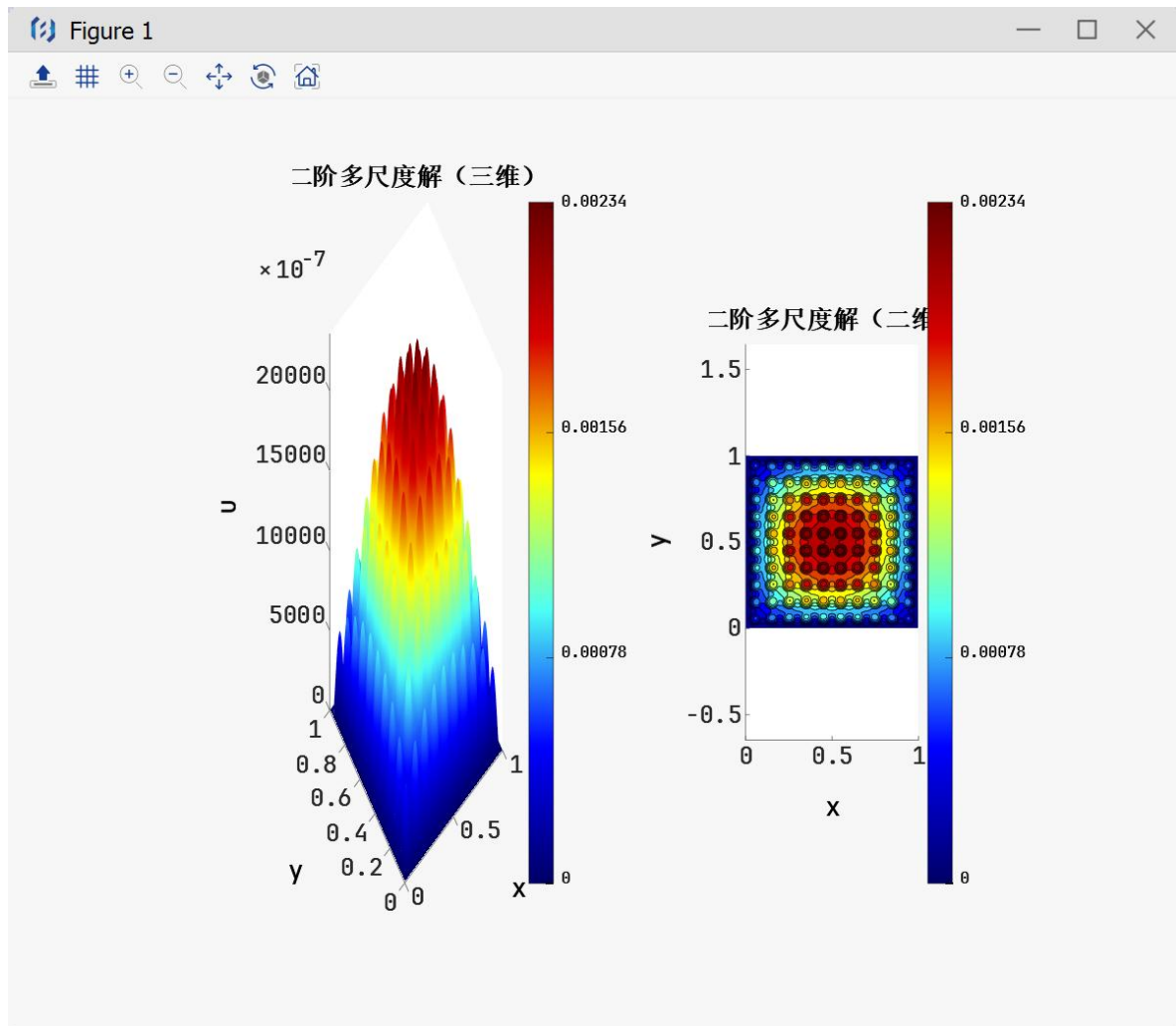
clc,clear;
%设定求解区域
xyt_range=[0,1,0,1,0,0.1];
%设定差分网格
Nmmt=[101,81,40,401];
%设定方程参数
equ_para.d_in=1;
equ_para.d_ma=1;
equ_para.c_in=1;
equ_para.c_ma=100;
equ_para.f=@(x,y,t)1;
equ_para.len=0.04;
equ_para.epsilon=0.1;
%设定初始条件
u0=@(x,y)0;
%设定边界条件
bc_type=[1,1,1,1];
bc_para.h_left=@(x,y,t)1;

```

```

bc_para.r_left=@(x,y,t)0;
bc_para.h_right=@(x,y,t)1;
bc_para.r_right=@(x,y,t)0;
bc_para.h_top=@(x,y,t)1;
bc_para.r_top=@(x,y,t)0;
bc_para.h_bottom=@(x,y,t)1;
bc_para.r_bottom=@(x,y,t)0;
%设定夹杂相材料形状
shape='circle';
solve_mulpur(xyt_range,Nmmtpe,equ_para,u0,bc_type,bc_para,shape)

```



六、多尺度双曲型偏微分方程（二阶多尺度解）

```

clc,clear;
%设定求解区域
xyt_range=[0,1,0,1,0,0.1];
%设定差分网格
Nmmtpe=[40,20,100,100];
%设定方程参数
equ_para.c_in=1;

```

```

equ_para.c_ma=100;
equ_para.d_in=1;
equ_para.d_ma=1;
equ_para.f=@(x,y,t)1;
equ_para.epsilon=0.1;
equ_para.len=0.04;
%设定初始条件
init.u0=@(x,y)0;
init.v0=@(x,y)0;
%设定边界条件
bc_type=[1,1,1,1];
bc_para.h_left=@(x,y,t)1;
bc_para.r_left=@(x,y,t)0;
bc_para.h_right=@(x,y,t)1;
bc_para.r_right=@(x,y,t)0;
bc_para.h_bottom=@(x,y,t)1;
bc_para.r_bottom=@(x,y,t)0;
bc_para.h_top=@(x,y,t)1;
bc_para.r_top=@(x,y,t)0;
%设定夹杂相材料形状
shape='circle';
solve_mulhyp(xyt_range,Nmmtp,equ_para,init,bc_type,bc_para,shape)

```

